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TITLE PAGE

Name of Department : IT Subject : DBMS

Experiment No. 10 Roll No. 36

Name of Expt. Write & Execute suitable databases Triggers.

Date of Performance of Experiment :- 10/5/23

Date of Completion of Experiment :- 15/5/23

Progressive Assessment Marks

Marks Obtained				Dated Sign Teacher
Attendance (3 Months)	Performance (4 Marks)	Writing & Submission (3 Marks)	Total (10 Marks)	
3	3	3	9	

* Practical No. 10 *

Title : Write and execute suitable database triggers.

Objectives :

Write and execute suitable database triggers. Consider now row level and statement level triggers.

Theory :

* Trigger -

A trigger is a stored procedure in database which automatically invokes whenever a special event in the database occurs. For example, a trigger can be invoked when a row is inserted into a specified table or when certain table columns are being updated. Trigger is EVENT-CONDITION-ACTION Model.

Example :

To create - Insert - Select Queries using Live Oracle SQL.

Similarly Trigger program is executed using LIVE ORACLE - SQL.

Page No.

Select * from customers ;

ID	NAME	AGE	ADDRESS	SALARY
1	Ramesh	32	Ahmedabad	2000
2	Khilan	25	Delhi	1500
3	Kaushik	23	Kota	2000
4	chaitali	25	Mumbai	6500
5	Hardik	27	Bhopal	8500
6	Komal	22	MP	4500

```
CREATE or REPLACE TRIGGER display_salary_changes
BEFORE DELETE or INSERT or UPDATE ON
customers FOR EACH ROW
WHEN (NEW.ID) >0
DECLARE
 sal_diff number ;
BEGIN
 sal_diff := : NEW.salary - : OLD.salary ;
dbms_output.put_line ('Old Salary :' || : OLD.
salary);
dbms_output.put_line ('New salary :' || :NEW.
salary);
dbms_output.put_line ('Salary difference :' ||
sal_diff);
END;
```

When the above code is executed at the SQL Prompt, it produce the following result -

Trigger created.

```
UPDATE customers
SET salary = salary + 500
WHERE id = 2;
```

When a record is updated in the CUSTOMERS table, the above create trigger, display_salary_changes will be fired and it will display the following result :

Old salary : 1500
New salary : 2000
Salary difference : 500

* Conclusion -

Thus, We successfully implemented the trigger query using sql.

```
Enter password: *************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 35
Server version: 8.0.32 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| customers          |
| employee           |
| information_schema |
| management         |
| medicine           |
| mysql              |
| performance_schema |
| student            |
| students           |
+--------------------+
9 rows in set (0.00 sec)

mysql> use students;
Database changed
mysql> select *from stud;
+----+---------+------+------------+
| id | Name    | Age  | Address    |
+----+---------+------+------------+
|  1 | Naz     |   20 | Shrirampur |
|  2 | Radhika |   19 | Loni       |
|  4 | Ram     |   24 | loni       |
+----+---------+------+------------+
3 rows in set (0.00 sec)


mysql> desc stud;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| id      | int         | NO   | PRI | NULL    |       |
| Name    | varchar(20) | YES  |     | NULL    |       |
| Age     | int         | YES  |     | NULL    |       |
| Address | varchar(30) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql> delete from stud;
Query OK, 3 rows affected (0.00 sec)

mysql> desc stud;
```

```
+---------+-------------+------+-----+---------+-------+
| Field     | Type               | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| id          | int                 | NO    | PRI | NULL      |          |
| Name      | varchar(20) | YES   |       | NULL      |          |
| Age        | int                 | YES   |       | NULL      |          |
| Address | varchar(30) | YES   |       | NULL      |          |
+---------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql> select *from stud;
Empty set (0.00 sec)

mysql> delimiter @@
mysql> create trigger newly
    ->        after update on stud
    ->        for each row
    ->        begin
    ->        if new.age <20 then
    ->        insert into stud values(new.id,2);
    ->        end if;
    ->        end;
    ->        @@
Query OK, 0 rows affected (0.01 sec)

mysql> select *from stud;
    -> @@
Empty set (0.01 sec)

mysql> update stud set age=15 where id=1;
    -> @@
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0   Changed: 0   Warnings: 0

mysql> select *from stud @@
+----+---------+------+------------+
| id | Name      | Age  | Address     |
+----+---------+------+------------+
|  1 | Naz        |   15 | Shrirampur |
|  2 | Radhika |   19 | Loni           |
|  4 | Ram       |   24 | loni            |
+----+---------+------+------------+
3 rows in set (0.00 sec)
```